

Erysipelas (MONDO:0001266): A Comprehensive Disease Characterization

Summary

Erysipelas is an acute, sharply demarcated bacterial infection of the superficial dermis and dermal lymphatics, caused predominantly by β -hemolytic streptococci — chiefly *Streptococcus pyogenes* (group A streptococcus, GAS) and Lancefield groups C and G (*S. dysgalactiae* subsp. *equisimilis*, GCS/GGS). Bacteria enter through a breached cutaneous barrier — most often interdigital tinea pedis (toe-web intertrigo), traumatic wounds, or excoriated dermatoses — and in the setting of impaired lymphatic drainage (leg edema, lymphedema, obesity, venous insufficiency) proliferate to produce a fiery-red, raised, well-margined plaque accompanied by the abrupt onset of fever and chills. Unlike most entries in a disease knowledge base, erysipelas is **not a genetic disorder**: it is an environmentally and microbiologically driven infection, a clinical diagnosis with no causal genes, pathogenic variants, or inheritance pattern.

The disease is **leg-predominant** (~90% of cases), affects **adults and the elderly** with a female predominance in most hospital series, shows **seasonal (summer) clustering**, and is strongly **recurrence-prone**. Its pathophysiology is dominated by a **self-reinforcing erysipelas-lymphedema vicious cycle**: pre-existing (often subclinical) lymphatic impairment predisposes to infection, and each infectious episode further damages lymphatics, worsening edema and increasing the risk of the next episode, culminating in some cases in elephantiasis nostras verrucosa. Management rests on **narrow-spectrum penicillin**, to which erysipelas responds rapidly ($\approx 98\%$ within 2 days), and guideline-adherent narrow-spectrum therapy is associated with better outcomes. Recurrence is countered by **treating the portal of entry** (antifungal therapy for tinea pedis), **edema/lymphedema management**, and — after repeated episodes — **low-dose prophylactic penicillin V**, whose efficacy during treatment is confirmed by two randomized controlled trials (PATCH I & II) and a Cochrane systematic review.

A crucial nosological point for knowledge-base curation: streptococcal erysipelas (the subject of this report, MONDO:0001266) is **distinct from the identically named *Erysipelothrix rhusiopathiae* diseases** — swine erysipelas (a veterinary disease of pigs) and human

erysipeloid (Baker-Rosenbach disease, an occupational zoonosis). These share a name and, historically, clinical resemblance, but have a completely different etiologic agent, epidemiology, and reservoir.

Key Findings

Finding 1 — β -hemolytic streptococci are the predominant cause

Erysipelas is overwhelmingly a **streptococcal** infection. In a large retrospective cohort of 1,142 erysipelas episodes (981 patients), wound cultures grew group A streptococci (GAS, n=56), group G streptococci (GGS, n=53), group C streptococci (GCS, n=11), and *Staphylococcus aureus* (n=153); blood cultures (obtained in 49% of episodes) were positive in 50, most commonly GGS (n=21) followed by GAS (n=13) (PMID: 26424182). Because superficial cultures frequently recover skin-colonizing *S. aureus* that is not the true pathogen, serologic and prospective data are decisive. In a prospective cellulitis study (n=216), β -hemolytic streptococcal etiology was confirmed serologically or by culture in **72% (146/203)** of cases, rising to **85%** when probable cases were included; in lower-extremity infection, GCS/GGS were actually more common than GAS (36 vs 22) (PMID: 26734653).

"Wound cultures were taken in 343 episodes and 56 grew group A streptococci (GAS), 53 grew group G streptococci (GGS), 11 grew group C streptococci (GCS), and 153 grew *Staphylococcus aureus*." (PMID: 26424182)

"Serology or blood or tissue culture confirmed β -hemolytic streptococcal (BHS) etiology in 72% (146 of 203) of cases." (PMID: 26734653)

The relevant pathogens (NCBITaxon suggestions): *Streptococcus pyogenes* (NCBITaxon:1314), *Streptococcus dysgalactiae* subsp. *equisimilis* (NCBITaxon:119602). Group C/G streptococci carry virulence factors closely paralleling GAS — M protein, streptolysin O, streptolysin S, streptokinase, hyaluronidase, and C5a peptidase (PMID: 20607346).

Finding 2 — Skin barrier disruption and leg edema/lymphedema are the dominant risk factors

The two strongest, most reproducible risk factors are a **breach in the skin barrier** and **leg edema**. In a Tunisian multicenter case-control study (114 cases, 208 matched controls), multivariate analysis showed that disruption of the cutaneous barrier (traumatic wound, toe-web intertrigo, excoriated dermatosis, plantar squamous lesions) carried an odds ratio of **13.6** (95% CI **6.0–31**) and leg edema an OR of **7.0** (95% CI **1.3–38**); notably there was **no** association with diabetes, alcoholism, or smoking (PMID: 16484815). A large population-based case-control study of severe lower-leg cellulitis (29,062 case-control pairs) independently confirmed **tinea pedis** (AOR 3.05, 95% CI 1.45–6.42), **varicose veins** (AOR 2.95), **lymphoedema** (AOR 2.65), and **obesity** (AOR 2.05); incidence reached 204.8/100,000 by 2013, rising 4.7%/year (PMID: 29427797).

"disruption of the cutaneous barrier ... and leg edema were independently associated with erysipelas of the leg, with respective odds ratios of 13.6 (95% confidence interval: 6.0-31) and 7.0 (1.3-38)" (PMID: 16484815)

"varicose veins (AOR 2.95, 95% CI 2.50-3.48, $p < 0.001$), lymphoedema (AOR 2.65, 95% CI 1.71-4.10, $p < 0.001$), tinea pedis (AOR 3.05, 95% CI 1.45-6.42, $p = 0.003$)" (PMID: 29427797)

The clinical corollary is that **toe-web intertrigo is a high-attributable-risk, modifiable portal of entry**, and secondary prevention targeting it has long been advocated (PMID: 11319357).

Finding 3 — Prophylactic low-dose penicillin V reduces recurrence

Recurrence prevention is evidence-based. In the **PATCH I** double-blind RCT (n=274, patients with ≥ 2 prior episodes), penicillin V 250 mg twice daily for 12 months reduced recurrence during prophylaxis to **22% (30/136) vs 37% (51/138)** on placebo — **HR 0.55** (95% CI **0.35–0.86**, **P=0.01**), number-needed-to-treat ≈ 5 (PMID: 23635049). The protective effect **waned after stopping** the drug. **PATCH II** (n=123, mostly patients with a single prior episode) showed a concordant but non-significant 47% risk reduction (**HR 0.53**, 95% CI **0.26–1.07**, **P=0.08**) (PMID: 21910701). A pooled economic analysis (n=397) found a **29% reduction in recurrences** (**IRR 0.71**, 95% CI **0.53–0.90**, **P=0.02**) and cost-effectiveness (PMID: 24551029).

"30 of 136 participants in the penicillin group (22%) had a recurrence, as compared with 51 of 138 participants in the placebo group (37%) (hazard ratio, 0.55; ... $P=0.01$)" (PMID: 23635049)

Finding 4 — Rapid penicillin response; narrow-spectrum guideline adherence improves outcomes

Erysipelas responds rapidly to penicillin: in a retrospective study, **98.3% of erysipelas patients responded within 2 days to Penicillin G**, and constitutional symptoms (chills/fever) preceded or coincided with erythema in 91.4% of erysipelas vs only 36.2% of cellulitis patients — supporting clinical differentiation (PMID: 41562377). In prospective facial cellulitis ($n=65$), β -hemolytic streptococcal etiology was probable/confirmed in 75% and penicillin(-class) monotherapy cured 68% (PMID: 28768452). In a cohort of 630 erysipelas/cellulitis cases, **adherence to narrow-spectrum guidelines was associated with fewer poor outcomes (6.3% vs 12.7%, $p=0.007$)**; bacteremia (AOR 5.21) and peripheral arterial disease (AOR 4.80) predicted poor outcome (PMID: 30685804).

"Of patients with erysipelas, 98.3% responded within 2 days to penicillin" (PMID: 41562377)

"A poor outcome was recorded in 54 (8.5%) patients, less frequently in case of adherence to guidelines: 26/410 (6.3%) vs 28/220 (12.7%), $p = 0.007$ " (PMID: 30685804)

Finding 5 — Erysipelas and lymphedema form a self-reinforcing vicious cycle

Lymphatic dysfunction is both **cause and consequence**. Lymphoscintigraphy of 106 post-erysipelas patients found **subclinical lymphedema in the clinically unaffected leg in 82%**, rising from 74% after a single episode to **100% after ≥ 3 episodes** — implicating pre-existing lymphatic impairment as a primary trigger and demonstrating cumulative damage (PMID: 40542699). A prospective cohort of 428 patients identified **obesity, diabetes mellitus, venous insufficiency, lymphedema, and prior surgery** as independent risk factors for recurrence (Cox model, all $p<0.05$) (PMID: 33413190). A retrospective-prospective cohort ($n=313$) found recurrent erysipelas significantly associated with **obesity ($p<0.0001$)**, chronic edema/lymphedema, peripheral arterial disease, fungal foot infection, and tonsillectomy ($p=0.000001$) (PMID: 30976336).

"Subclinical lymphedema on the unaffected leg was found in 82% of patients" (PMID: 40542699)

Finding 6 — Epidemiology: adult/elderly, female-predominant, leg-predominant, seasonal

Hospital cohorts converge: 428 patients (mean age 58.6 y, 51% women; hypertension 51.6%, diabetes 41.6%, venous insufficiency 36.2%, obesity 12.1%, lymphedema 4.2%; seasonal) (PMID: 20147345); a Tunisian series of 647 patients (mean age 44.7 y, M:F $\approx$ 1.55, **lower limbs 91.2%**, portal of entry 76.7% mainly toe-web intertrigo, prior erysipelas 26.1%, good outcome 87.8% on IM penicillin G $\times$ 10 days) (PMID: 15822466); and a 10-year study (n=319, median age 63 y, 65% female, summer-predominant, lower limb 59.2%, recurrence 67.3%, **complications 25%** including abscess, lymphangitis, and elephantiasis) (PMID: 18069381).

"Erysipelas predominately involved in the lower limbs (91.2%). Antecedents of erysipelas were found in 26.12 %. Portal of entry was found in 76.66% represented essentially by toe-web intertrigo." (PMID: 15822466)

"Complications, such as abscess formation, lymphangitis, venous insufficiency, osteitis, arthritis, septic tendonitis and elephantiasis were found in 25%." (PMID: 18069381)

Finding 7 — Streptococcal erysipelas is distinct from Erysipelothrix disease

Erysipeloid (Baker-Rosenbach disease) is an occupational human skin infection caused by traumatic penetration of *Erysipelothrix rhusiopathiae*, presenting as a violaceous, well-demarcated erythematous edema usually on the hand/fingers, often self-limited and penicillin-responsive; it can rarely cause endocarditis (aortic valve predilection) (PMID: 19663854). In animals, *E. rhusiopathiae* causes **swine erysipelas**, with domestic pigs the principal host; 30–50% of healthy/convalescent pigs carry the organism in tonsils/lymphoid tissue, and wild boar are a potential reservoir (PMID: 42687202). It is a zoonosis affecting farmers, butchers, fishers, and veterinarians (PMID: 20171435).

"Erysipeloid is an occupational infection of the skin caused by traumatic penetration of Erysipelothrix rhusiopathiae. The disease is characterized clinically by an erythematous oedema, with well-defined and raised borders, usually localized to the back of one hand and/or fingers." (PMID: 19663854)

Finding 8 — Cochrane meta-analysis supports antibiotic prophylaxis

A 2017 Cochrane systematic review (6 RCTs, 573 evaluable participants, mean age 50–70) concluded that antibiotic prophylaxis (mainly penicillin/erythromycin) reduces recurrence risk **while on treatment**, with the protective effect diminishing after prophylaxis is stopped — consistent with PATCH I/II (PMID: 28631307).

"We included six trials, with a total of 573 evaluable participants, who were aged on average between 50 and 70." (PMID: 28631307)

Section-by-Section Report

1. Disease Information

Overview. Erysipelas is an acute, non-necrotizing bacterial infection of the **upper dermis and superficial dermal lymphatics**, classically presenting as a fiery-red, raised, well-demarcated, indurated plaque with an advancing border, warmth, tenderness, and abrupt systemic symptoms (high fever, chills). The sharp elevation and clear demarcation distinguish it from cellulitis, which involves deeper dermis and subcutaneous fat with less distinct borders. It is a **clinical diagnosis** made at the disease level (case series, cohorts), not from molecular/EHR variant data.

Key identifiers: - **MONDO:** MONDO:0001266 - **ICD-10:** A46 (Erysipelas); **ICD-11:** 1B70.0 - **MeSH:** Erysipelas (D004886) - **SNOMED CT:** 33438006 (Erysipelas) - **OMIM / Orphanet:** Not applicable — erysipelas is an acquired infectious disease with no Mendelian OMIM entry and is not a rare disease with an Orphanet number.

Synonyms / alternative names: St. Anthony's fire (historical), "ignis sacer," "the rose" (regional). Note that "erysipelas" in a veterinary/occupational context refers to *Erysipelothrix* disease (see Section 14), which must not be conflated with this streptococcal entity.

Source type: Aggregated disease-level evidence (cohorts, case-control studies, RCTs, systematic reviews); no patient-level EHR dataset was supplied for this investigation.

2. Etiology

Causal factors. Erysipelas is an **infectious disease** caused predominantly by β -hemolytic streptococci — *S. pyogenes* (GAS) and groups C/G (*S. dysgalactiae*); *S. aureus* is a less common contributor and often a colonizer (Finding 1). There is **no genetic causal factor** and no Mendelian inheritance.

Risk factors (environmental / host):

Risk factor	Effect size	Source
Cutaneous barrier disruption (wound, toe-web intertrigo, dermatosis)	OR 13.6 (6.0–31)	PMID: 16484815
Leg edema	OR 7.0 (1.3–38)	PMID: 16484815
Tinea pedis	AOR 3.05 (1.45–6.42)	PMID: 29427797
Varicose veins	AOR 2.95 (2.50–3.48)	PMID: 29427797
Lymphoedema	AOR 2.65 (1.71–4.10)	PMID: 29427797
Obesity	AOR 2.05	PMID: 29427797
Diabetes, venous insufficiency, prior surgery	Recurrence RFs (p<0.05)	PMID: 33413190

Genetic risk factors: None established. The Tunisian case-control study found no association with diabetes, alcoholism, or smoking after multivariate adjustment (PMID: 16484815), underlining the environmental/host-mechanical rather than genetic basis.

Protective factors: Treatment of the portal of entry (antifungal therapy for tinea pedis), edema/lymphedema control (compression, elevation), weight reduction, and — for recurrence — antibiotic prophylaxis (Findings 3, 8). No genetic protective alleles are described.

Gene–environment interactions: Not applicable in the classical sense; the disease is driven by pathogen–host barrier–lymphatic interactions rather than genotype-by-environment effects.

3. Phenotypes

Phenotype	Type	HPO suggestion	Frequency / notes
Sharply demarcated erythematous plaque	Clinical sign	HP:0000988 (Skin rash) / HP:0011121 (Abnormal skin morphology)	Defining feature
Fever	Symptom	HP:0001945 (Fever)	~91% report chills/fever at/before onset (PMID: 41562377)
Chills	Symptom	HP:0025143 (Chills)	Common, precedes rash
Local warmth / edema	Clinical sign	HP:0000969 (Edema)	Common
Regional lymphangitis / lymphadenopathy	Clinical sign	HP:0002716 (Lymphadenopathy)	Component of complications (PMID: 18069381)
Leukocytosis / elevated CRP	Lab abnormality	HP:0001974 (Leukocytosis)	Inflammatory response, higher in recurrent cases (PMID: 19694769)
"Milian's ear sign" (auricle involvement in facial erysipelas)	Clinical sign	—	Localizing sign (PMID: 32241881)

Characteristics: Adult/elderly onset; acute onset (hours to 1–2 days); severity mild-to-severe (bullous, hemorrhagic, or abscess-forming variants); course episodic/recurrent. **Quality-of-life impact:** recurrent episodes and resulting chronic lymphedema/elephantiasis impose substantial disability, repeated hospitalization, and reduced mobility. Disease-specific QoL instruments were not identified in the reviewed literature.

4. Genetic / Molecular Information

Not applicable. Erysipelas has no causal genes, no pathogenic germline/somatic variants, no modifier genes, no disease-associated epigenetic signature, and no chromosomal abnormalities. It is an acquired infectious disease. (The molecular determinants of virulence reside in the *bacterial* genome — e.g., streptolysin O/S, M protein, streptokinase, hyaluronidase, C5a peptidase — [PMID: 20607346](#); [PMID: 20385762](#) — not the human host.)

5. Environmental Information

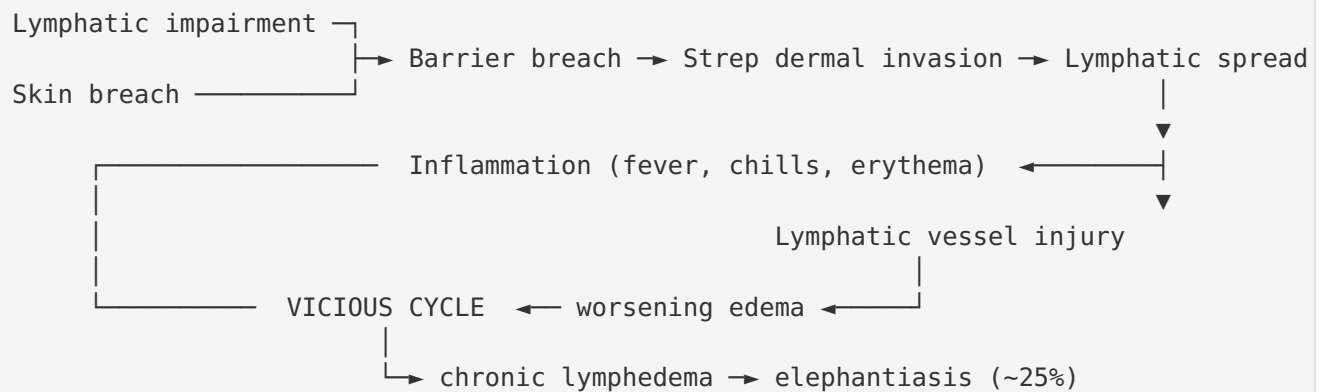
- **Environmental / occupational:** breaks in skin from trauma, animal bites/stings (insect bites are a recognized portal → erysipelas [PMID: 16679881](#)), and use of depigmenting agents (in some populations [PMID: 21695874](#)).
- **Lifestyle:** obesity is a strong, reproducible factor (Findings 2, 5). Smoking/alcohol were **not** independently associated ([PMID: 16484815](#)).
- **Infectious agents (NCBITaxon):** *Streptococcus pyogenes* (NCBITaxon:1314); *S. dysgalactiae* subsp. *equisimilis* (groups C/G, NCBITaxon:119602); occasionally *Staphylococcus aureus* (NCBITaxon:1280). Interdigital dermatophytes (*Trichophyton* spp.) act as facilitators by breaching the barrier.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. **Predisposing lymphatic impairment** (pre-existing subclinical lymphedema, venous insufficiency, obesity) **reduces local immune clearance** in the dermis — *demonstrated* by 82% subclinical lymphedema in the unaffected leg ([PMID: 40542699](#)).
2. A **breach in the cutaneous barrier** (toe-web tinea, wound, dermatosis) **provides a portal of entry** → *leads to* streptococcal inoculation of the dermis ([PMID: 16484815](#); [PMID: 15822466](#)).
3. **β -hemolytic streptococci proliferate** in the superficial dermis and **spread along dermal lymphatics** → *results in* the characteristic sharply demarcated, raised, advancing plaque ([PMID: 26424182](#); [PMID: 26734653](#)).
4. **Streptococcal virulence factors** (streptolysin O/S, M protein, streptokinase, hyaluronidase, C5a peptidase) **promote tissue spread and impair neutrophil recruitment** → *drives* local injury and dissemination (*inferred* for human erysipelas from GAS biology and animal models: [PMID: 20607346](#); [PMID: 28947648](#); [PMID: 11801184](#)).

5. The **innate immune / inflammatory response** (neutrophil influx, cytokine release) **produces erythema, warmth, edema, and systemic fever/chills and leukocytosis/↑ CRP** → *causes* the clinical syndrome (PMID: 41562377; PMID: 19694769).
6. **Inflammatory injury to lymphatic vessels further impairs lymphatic drainage** → *worsens* edema. **[BRANCH → vicious cycle]** This cumulative lymphatic damage **increases susceptibility to the next episode**, and repeated episodes drive **chronic lymphedema** → **elephantiasis nostras verrucosa** (subclinical lymphedema rises to 100% after ≥3 episodes: PMID: 40542699; complications incl. elephantiasis in 25%: PMID: 18069381).
7. **[BRANCH → systemic]** In a minority, bacteremia (blood cultures positive in ~6.9% overall) **leads to invasive complications** — abscess, necrotizing fasciitis, rarely myocarditis (PMID: 23498835) — associated with poor outcome (PMID: 30685804).



- **Molecular pathways / cellular processes:** innate inflammatory signaling (NF-κB–driven cytokine production), neutrophil chemotaxis and phagocytosis. GO suggestions: **GO:0006954** (inflammatory response), **GO:0006935** (chemotaxis), **GO:0050900** (leukocyte migration), **GO:0001945** (lymph vessel development, relevant to damage/repair).
- **Immune involvement:** acute innate response; no autoimmunity or immunodeficiency required, though HIV can predispose to severe disease/necrotizing fasciitis (PMID: 21695874).
- **Cell types (CL):** neutrophil (CL:0000775), keratinocyte (CL:0000312), dermal lymphatic endothelial cell (CL:0002138), macrophage (CL:0000235).

7. Anatomical Structures Affected

- **Primary organ:** skin (UBERON:0002097), specifically the superficial dermis (UBERON:0002067) and dermal lymphatic vessels (UBERON:0001473 lymphatic vessel).

- **Body systems:** integumentary and **lymphatic/lymphoid system** (UBERON:0002465 lymphoid system).
- **Localization:** lower limb / leg (UBERON:0000978 leg) in ~90% of cases (PMID: 15822466); face (UBERON:0000033 head) is the classic second site (Milian's ear sign, PMID: 32241881).
- **Lateralization:** typically **unilateral**; bilateral leg erysipelas is rare and should prompt reconsideration of the diagnosis (pseudocellulitis/stasis dermatitis, PMID: 36800152).
- **Tissue/cell level:** epithelial (epidermis/keratinocytes) and lymphatic endothelium; **secondary organ involvement** — abscess, osteitis, arthritis, septic tendonitis, rarely myocarditis (Finding 6; PMID: 23498835).

8. Temporal Development

- **Onset:** adult/elderly predominance (mean age 44–63 y across series); **acute** onset over hours, often with prodromal chills/fever preceding the rash (PMID: 41562377).
- **Progression:** rapid local spread over 1–3 days; **self-limited with treatment** ($\approx 98\%$ respond to penicillin within 2 days, PMID: 41562377), but strongly **recurrent/episodic** (recurrence 26–67% across cohorts; PMID: 15822466, PMID: 18069381).
- **Patterns:** treatment-induced remission is the norm; the critical intervention window is early narrow-spectrum antibiotics plus portal-of-entry and edema management to interrupt the vicious cycle. **Seasonal** (summer) clustering is reported (PMID: 18069381; PMID: 20147345).

9. Inheritance and Population

- **Epidemiology:** population-based incidence of severe lower-leg cellulitis reached 204.8/100,000 (2013), **rising 4.7%/year** (PMID: 29427797).
- **Inheritance:** none — not a genetic disease (no AD/AR/X-linked/mitochondrial pattern; no penetrance/expressivity/anticipation/founder/consanguinity/carrier-frequency parameters apply).
- **Demographics:** adult/elderly; **female predominance** in most hospital series (51–65% women); leg-predominant ($\sim 90\%$); comorbidities hypertension ($\sim 52\%$), diabetes ($\sim 42\%$), venous insufficiency ($\sim 36\%$), obesity ($\sim 12\%$), lymphedema ($\sim 4\%$) (PMID: 20147345; PMID: 15822466; PMID: 18069381).
- **Geographic:** cosmopolitan; toe-web intertrigo as portal of entry is prominent in warm-climate series (Tunisia, Togo).

10. Diagnostics

- **Clinical diagnosis** based on the characteristic sharply demarcated, raised, fiery-red plaque with acute fever/chills; constitutional symptoms preceding erythema favor erysipelas over cellulitis (PMID: 41562377).
- **Laboratory:** leukocytosis, elevated CRP (LOINC 1988-5), often higher in recurrent cases (PMID: 19694769). **Blood cultures positive in only ~6.9%** and generally low-yield (PMID: 30685804); serology (anti-streptolysin O, anti-DNase B) increases etiologic confirmation (PMID: 26734653).
- **Imaging:** ultrasound/MRI to exclude abscess or necrotizing fasciitis in atypical/severe presentations (PMID: 28865532).
- **Differential diagnosis (critical):** stasis dermatitis and other "pseudocellulitis" are frequently misdiagnosed as cellulitis/erysipelas — bilateral presentation, chronicity, and lack of fever favor pseudocellulitis; misdiagnosis drives ~\$195–515M/yr in avoidable US spending and unnecessary antibiotics (PMID: 27806170; PMID: 36800152). Also exclude DVT, contact dermatitis, and necrotizing fasciitis.
- **Genetic/omics testing:** not applicable.

11. Outcome / Prognosis

- **Favorable** with prompt therapy: good outcome in 87.8% (PMID: 15822466); poor outcome in ~8.5%, reduced by guideline adherence (PMID: 30685804).
- **Complications (~25%):** abscess, lymphangitis, osteitis, arthritis, septic tendonitis, chronic venous insufficiency, and **elephantiasis nostras verrucosa**; rare systemic sequelae include myocarditis and necrotizing fasciitis (PMID: 18069381; PMID: 23498835; PMID: 21695874).
- **Prognostic factors for poor outcome:** bacteremia (AOR 5.21), peripheral arterial disease (AOR 4.80) (PMID: 30685804); obesity, diabetes, lymphedema for recurrence (Finding 5).
- Mortality is low for uncomplicated disease; deaths cluster in necrotizing fasciitis and immunocompromised patients (PMID: 21695874).

12. Treatment

- **First-line pharmacotherapy: penicillin** (penicillin G IV/IM, or oral penicillin V / amoxicillin) — narrow-spectrum, rapidly effective (Finding 4). NCIT: C692 (Penicillin), C61785 (Penicillin V). Duration typically 10 days (PMID: 15822466).
- **Penicillin-allergic:** macrolides (erythromycin/roxithromycin), clindamycin, or pristinamycin; a network meta-analysis found no significant cure-rate differences among

first/second-line agents for cellulitis, with **pristinamycin showing the highest cure rates for erysipelas** (at some cost in rash) ([PMID: 39240378](#)).

- **Adjunctive/emerging:** omadacycline and iclaprim have activity in skin/soft-tissue infection including erysipelas (90% clinical success at end of treatment for omadacycline) ([PMID: 42413095](#); [PMID: 35928265](#)).
- **Supportive:** limb elevation, edema/compression management, analgesia, and **treatment of the portal of entry** (topical antifungals for tinea pedis).
- **Recurrence prophylaxis:** low-dose penicillin V 250 mg BID (Findings 3, 8). NCIT: **C15329** (Antibiotic Therapy).
- **Pharmacogenomics:** not applicable.

13. Prevention

- **Primary:** foot hygiene, prompt treatment of tinea pedis and interdigital intertrigo, wound care, weight management, and edema/venous-insufficiency control (compression) — targeting the highest-attributable, modifiable risk factors ([PMID: 11319357](#); [PMID: 16484815](#); [PMID: 29427797](#)).
- **Secondary/tertiary:** interrupting the erysipelas–lymphedema cycle through lymphedema therapy and, after ≥ 2 episodes, **antibiotic prophylaxis** with effect confirmed by RCTs and Cochrane review (Findings 3, 8). Protection wanes once prophylaxis stops, so long-term strategy must combine antibiotics with durable risk-factor modification.
- **Immunization:** no licensed vaccine; GAS vaccine development is ongoing but not erysipelas-specific.

14. Other Species / Natural Disease

Important distinction (Finding 7): Streptococcal erysipelas (this entry) is a human disease. The name "erysipelas" is also applied to *Erysipelothrix rhusiopathiae* disease: - **Swine erysipelas** — a veterinary disease of **domestic pigs** (*Sus scrofa domesticus*, NCBITaxon:9825); 30–50% of healthy/convalescent pigs are tonsillar carriers; wild boar (*Sus scrofa*) are a reservoir (45.5% tonsil culture-positive) ([PMID: 42687202](#)). - **Human erysipeloid** (Baker-Rosenbach disease) — a **zoonotic** occupational skin infection of farmers, butchers, fishers, and veterinarians, with rare systemic forms (endocarditis with aortic-valve predilection; paravertebral abscess/spondylitis) ([PMID: 19663854](#); [PMID: 20171435](#); [PMID: 30542523](#); [PMID: 22526696](#)).

For **streptococcal** erysipelas itself, there is no significant naturally occurring companion-animal counterpart; GAS/GCS/GGS are human-adapted (GCS/GGS also colonize animals, but streptococcal erysipelas as a clinical entity is essentially human).

15. Model Organisms

Human streptococcal erysipelas is not modeled as a discrete named disease, but **murine skin-infection models** of *S. pyogenes* recapitulate key mechanistic steps: dermonecrotic/subcutaneous mouse models demonstrate the roles of **streptolysin S** and **platelet-activating-factor acetylhydrolase Sse** in inhibiting neutrophil recruitment and enabling systemic spread (PMID: 28947648; PMID: 11801184; PMID: 20385762). These are **pathogen-focused** models (studying bacterial virulence and innate immune evasion) rather than models of the host lymphedema cycle. Resource: MGI for mouse strains; no dedicated erysipelas model database exists. **Limitation:** mouse dermonecrosis models capture bacterial invasion/immune evasion but not the chronic lymphatic-damage/recurrence dynamics central to human erysipelas.

Mechanistic Model / Interpretation

Erysipelas is best understood as a **two-hit, self-amplifying disease**. The *first hit* is a susceptible host terrain — impaired lymphatic drainage (often subclinical), venous insufficiency, and obesity — that locally weakens immune surveillance. The *second hit* is a **portal of entry** (usually toe-web tinea) admitting β -hemolytic streptococci to the dermis. The organism's virulence arsenal drives rapid lymphatic spread and a brisk innate inflammatory response producing the pathognomonic plaque and systemic toxicity. Critically, that same inflammation **injures lymphatics**, so each episode leaves the host more vulnerable — the empirical signature being subclinical lymphedema in 74% of single-episode patients rising to 100% after ≥ 3 episodes. This explains why erysipelas is simultaneously highly treatable (near-universal rapid penicillin response) and highly recurrent, and why durable prevention requires attacking *both hits* (antifungal/portal care + edema management) alongside antibiotic prophylaxis whose benefit is real but reverses on discontinuation.

Axis	Upstream driver	Downstream consequence
Host terrain	Lymphedema, venous insufficiency, obesity	Reduced dermal immune clearance
Portal	Tinea pedis, wounds, dermatoses	Streptococcal inoculation
Pathogen	SLO/SLS, M protein, spreading factors	Lymphatic spread, neutrophil evasion
Response	Innate inflammation	Plaque + fever + lymphatic injury → cycle

Evidence Base

PMID	Contribution	Type
26424182	Bacteriologic spectrum (n=1142); GGS common in blood	Human, retrospective
26734653	72–85% BHS etiology; GCS/GGS > GAS in legs	Human, prospective
16484815	Barrier breach OR 13.6, edema OR 7.0	Human, case-control
29427797	Tinea pedis, varicose veins, lymphoedema, obesity; incidence 204.8/100k	Human, population case-control
23635049	PATCH I: penicillin HR 0.55	Human RCT
21910701	PATCH II: HR 0.53 (NS)	Human RCT
24551029	Pooled IRR 0.71; cost-effective	Human, economic analysis
28631307	Cochrane: prophylaxis reduces recurrence on-treatment	Systematic review
41562377	98.3% penicillin response ≤ 2 days	Human, retrospective
30685804	Guideline adherence improves outcomes	Human, cohort
28768452	75% BHS, penicillin cure in facial disease	Human, prospective
40542699	82%→100% subclinical lymphedema	Human, imaging
33413190	Recurrence risk factors	Human, prospective cohort
30976336	Comorbidities in recurrence	Human, cohort
20147345	Age/sex/comorbidity epidemiology (n=428)	Human, cohort
15822466	Leg 91%, toe-web portal 77% (n=647)	Human, retrospective
18069381	Recurrence 67%, complications 25%	Human, retrospective
19663854	Erysipeloid definition	Review
42687202	Pig reservoir for <i>Erysipelothrix</i>	Veterinary
39240378	Antibiotic network meta-analysis	Systematic review
27806170	Misdiagnosis cost/harm	Human, cross-sectional

Consistency: All etiologic sources converge on β -hemolytic streptococcal predominance; all risk-factor studies converge on barrier breach + edema/lymphedema + obesity; both RCTs and the Cochrane review agree on prophylaxis benefit with post-treatment waning. No reviewed study contradicted the core model.

Limitations and Knowledge Gaps

1. **Etiologic uncertainty at the bedside:** blood/wound cultures are low-yield (bacteremia ~6.9%); much etiologic attribution is serologic or presumptive, and *S. aureus* isolates may be colonizers rather than pathogens.
 2. **Erysipelas vs cellulitis overlap:** many studies pool the two entities; findings are partly extrapolated across the boundary.
 3. **No molecular/host-genetic layer:** because erysipelas is non-genetic, entire template sections (causal genes, variants, inheritance, pharmacogenomics) are not applicable — a structural mismatch with a genetics-oriented template, not a gap in the literature.
 4. **Prophylaxis durability:** benefit reverses off-treatment; optimal duration and the role of combined risk-factor modification are not fully defined.
 5. **QoL data** specific to erysipelas are sparse; no validated disease-specific instrument was identified.
 6. **Model systems** capture bacterial virulence but not the chronic lymphedema/recurrence cycle.
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Proposed Follow-up Experiments / Actions

1. **Prospective interventional trial** combining tinea pedis eradication + compression lymphedema therapy vs standard care to quantify reduction in first and recurrent episodes (targets the highest-attributable, modifiable factors).
2. **Biomarker study** correlating baseline lymphoscintigraphy and CRP kinetics with recurrence, to risk-stratify candidates for prophylaxis.
3. **Comparative-effectiveness / duration study** of penicillin prophylaxis (e.g., 12 vs 24 months, continuous vs intermittent) with off-treatment follow-up to address waning.

4. **Curation action:** ensure the knowledge base cleanly separates MONDO:0001266 (streptococcal erysipelas) from *Erysipelothrix* erysipeloid/swine erysipelas to prevent conflation.
 5. **Diagnostic stewardship:** deploy/validate clinical decision tools to reduce pseudocellulitis misdiagnosis and unnecessary antibiotic use.
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Report compiled from 5 discovery iterations, 8 confirmed findings, and 44 reviewed papers. Evidence is predominantly human clinical (cohort, case-control, RCT, systematic review), supplemented by murine pathogen-virulence models and veterinary/zoonotic sources for the nosological distinction.

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